

Appendix A: Supplementary data

Diet matrix for each functional group corresponding to the demersal food web in the San Matías gulf in 1970.

<i>Prey \ predator</i>	1	2	3	4	5	6	7	8	9	10	11	12
1 Dolphins	0	0	0	0	0	0	0	0	0	0	0	0
2 Sea Lions	0	0	0	0	0	0	0	0	0	0	0	0
3 Other rays	0	0	0	0	0	0	0	0	0	0	0	0
4 Large rays	0	0	0	0	0	0	0	0	0	0	0	0
5 Small rays	0	0.003	0	0	0	0	0.015	0	0	0.035	0	0
6 Large sharks	0	0	0	0	0	0	0	0	0	0	0	0
7 Medium-sized sharks	0	0	0	0	0	0	0.001	0	0	0	0	0
8 Juvenile sharks	0	0	0	0	0	0	0.017	0	0	0	0	0
9 Narrownose smooth-hound	0	0	0	0	0	0	0	0	0	0	0	0
10 Benthic-demersal fish II	0	0	0	0	0	0	0	0	0	0.002	0	0
11 Flounders	0	0.003	0	0.012	0	0	0.034	0	0.041	0.035	0.01	0
12 Benthic-demersal fish I	0	0.055	0.004	0.002	0.001	0.17	0.079	0.073	0.131	0.069	0.15	0.001
13 Plownose chimaera	0.037	0	0	0	0	0	0	0	0	0	0	0
14 Large hake	0	0.012	0	0	0	0	0	0	0	0	0	0
15 Medium-sized hake	0	0.271	0.018	0.461	0	0	0.032	0.009	0	0.032	0.025	0
16 Juvenile hake	0	0.087	0.018	0.11	0	0.151	0.021	0.009	0	0.025	0.025	0
17 Pink cusk-eel	0.03	0.089	0	0.069	0	0	0	0	0	0	0	0
18 Patagonian hoki	0	0.013	0	0	0	0	0	0	0	0.11	0	0
19 Medium-sized pelagic fishes	0	0.345	0	0	0	0	0.043	0.02	0.079	0.035	0.12	0
20 Small pelagic fishes	0.107	0.022	0.091	0	0	0.023	0.021	0.012	0.075	0.035	0.094	0.089
21 Squids	0.622	0.05	0.007	0.003	0.001	0.025	0.097	0.09	0.054	0.109	0.08	0.088
22 Benthic organisms	0.204	0	0.543	0.325	0.915	0.631	0.48	0.44	0.38	0.159	0.069	0.456
23 Jellyfishes	0	0	0.002	0	0.003	0	0.021	0	0.009	0	0.051	0.137
24 Zooplankton	0	0	0.007	0	0	0	0	0	0	0.041	0	0.059
25 Phytoplankton	0	0	0	0	0	0	0	0	0	0	0	0
26 Detritus	0	0	0.009	0	0	0	0	0	0	0	0	0.049
27 Import	0	0.049	0.301	0.017	0.08	0	0.139	0.347	0.23	0.315	0.376	0.122

Diet matrix (cont.) for each functional group corresponding to the demersal food web in the San Matías gulf in 1970.

<i>Prey \ predator</i>	13	14	15	16	17	18	19	20	21	22	23	24
1 Dolphins	0	0	0	0	0	0	0	0	0	0	0	0
2 Sea Lions	0	0	0	0	0	0	0	0	0	0	0	0
3 Other rays	0	0	0	0	0	0	0	0	0	0	0	0
4 Large rays	0	0	0	0	0	0	0	0	0	0	0	0
5 Small rays	0	0	0	0	0	0	0	0	0	0	0	0
6 Large sharks	0	0	0	0	0	0	0	0	0	0	0	0
7 Medium-sized sharks	0	0	0	0	0	0	0	0	0	0	0	0
8 Juvenile sharks	0	0	0	0	0	0	0	0	0	0	0	0
9 Narrownose smooth-hound	0	0	0	0	0	0	0	0	0	0	0	0
10 Benthic-demersal fish II	0	0	0	0	0	0	0	0	0	0	0	0
11 Flounders	0	0	0	0	0	0	0	0	0	0	0	0
12 Benthic-demersal fish I	0.076	0.078	0.022	0.014	0.001	0	0	0	0	0	0	0
13 Plownose chimaera	0.001	0	0	0	0	0	0	0	0	0	0	0
14 Large hake	0	0	0	0	0	0	0	0	0	0	0	0
15 Medium-sized hake	0	0.46	0.075	0	0	0	0	0	0	0	0	0
16 Juvenile hake	0	0.109	0.117	0.027	0.026	0.041	0	0	0	0	0	0
17 Pink cusk-eel	0	0	0	0	0.019	0	0	0	0	0	0	0
18 Patagonian hoki	0	0	0	0	0.054	0.005	0	0	0	0	0	0
19 Medium-sized pelagic fishes	0.106	0.241	0	0	0.001	0.031	0.048	0	0	0	0	0
20 Small pelagic fishes	0.109	0.081	0.617	0.384	0.003	0.031	0.091	0	0.063	0	0	0
21 Squids	0.051	0.022	0.055	0.009	0.047	0.082	0.01	0	0.015	0	0	0
22 Benthic organisms	0.613	0.009	0.063	0.031	0.507	0.356	0.02	0.001	0.078	0.021	0	0
23 Jellyfishes	0	0	0	0	0.004	0.02	0.081	0.135	0.012	0	0.008	0
24 Zooplankton	0	0.001	0.048	0.366	0.186	0.407	0.322	0.705	0.177	0	0.689	0.01
25 Phytoplankton	0	0	0	0	0	0.022	0	0.129	0	0	0.018	0.95
26 Detritus	0	0	0	0	0	0	0	0	0	0.874	0.018	0.04
27 Import	0.043	0	0.002	0.169	0.152	0.005	0.428	0.03	0.655	0.105	0.267	0

Index of data pedigree generated for each functional group based on the data quality of input parameters (Biomass (B, t.km⁻²), Production to Biomass (P/B, year⁻¹), Consumption to Biomass (Q/B, year⁻¹) and Catches (B, t.km⁻²) estimates). Key for generating the index provided.

Group name	Biomass	P/B	Q/B	Diet	Catch
Dolphins	4	5	4	6	5
Sea Lions	6	5	4	6	5
Other rays	4	4	4	3	
Large rays	4	5	4	3	5
Small rays	4	5	4	3	5
Large sharks	3	4	4	3	5
Medium-sized sharks	4	4	4	3	5
Juvenile sharks	4	4	4	3	5
Narrownose smooth-hound	5	5	4	3	5
Benthic-demersal fish II	5	4	4	5	5
Flounders	6	5	4	5	5
Benthic-demersal fish I	4	4	4	3	5
Plownose chimaera	5	4	4	6	5
Large hake	6	8	7	6	6
Medium-sized hake	6	8	7	6	6
Juvenile hake	5	8	7	6	6
Pink cusk-eel	5	5	4	3	5
Patagonian hoki	5	5	4	3	5
Medium-sized pelagic fishes	4	5	4	3	
Small pelagic fishes	5	5	4	3	1
Squids	4	5	4	4	
Benthic organisms	3	4	4	3	1
Jellyfishes	3	4	4	3	
Zooplankton	1	4	4	3	
Phytoplankton	1	4			

Key for generating the Pedigree index.

Index value	Biomass	P/B - Q/B	Diet	Catch
1	Estimated by Ecopath	Estimated by Ecopath	General knowledge of related group	Guesstimate
2	From other model	Guesstimate	From other model	From other model
3	Guesstimate	From other model	General knowledge for same group	FAO statistics
4	Approximate or indirect method	Empirical relationship	Qualitative diet composition study	Regional Statistics (RG)
5	Sampling locally, low precision	Similar species, similar system, low precision	Quantitative but limited diet composition	RG combined with local studies. Low precision
6	Sampling locally, high precision	Similar species, same system, low precision	Quantitative, detailed, diet composition study	RG combined with local studies. High precision
7	-	Same species, similar system, low precision	-	-
8	-	Same species, same system, low precision	-	-

Summary statistics and metrics corresponding to the demersal food web in the San Matías gulf in 1970. Asterisks indicate parameters that depend on the simulated catches introduced to fit the model.

<i>Parameter</i>	<i>Value</i>	<i>Units</i>
Sum of all consumption	815.426	t.km ⁻² .year ⁻¹
Sum of all exports	95.486	t.km ⁻² .year ⁻¹
Sum of all respiratory flows	353.474	t.km ⁻² .year ⁻¹
Sum of all flows into detritus	214.686	t.km ⁻² .year ⁻¹
Total system throughput	1479.074	t.km ⁻² .year ⁻¹
Sum of all production	674.487	t.km ⁻² .year ⁻¹
Mean trophic level of the catch *	4.054	
Gross efficiency (catch/net p.p.) *	4.750E-06	
Calculated total net primary production	375.620	t.km ⁻² .year ⁻¹
Total primary production/total respiration	1.062	
Net system production	22.146	t.km ⁻² .year ⁻¹
Total primary production/total biomass	5.642	
Total biomass/total throughput	0.045	year ⁻¹
Total biomass (excluding detritus)	66.568	t.km ⁻²
Total catch *	0.002	t.km ⁻² .year ⁻¹
Connectance Index	0.275	
System Omnivory Index	0.391	
Ecopath pedigree index	0.534	
Ascendency	2187.4	Flowbits
Overhead	4527.2	Flowbits
Capacity	6714.6	Flowbits
Ascendency (%)	32.6	
Overhead (%)	67.4	

[illegible]

Vulnerabilities matrix (cont.) used to fit the simulation between 1970 and 2009

Prey \ predator	13	14	15	16	17	18	19	20	21	22	23	24
1 Dolphins												
2 Sea Lions												
3 Other rays												
4 Large rays												
5 Small rays												
6 Large sharks												
7 Medium-sized sharks												
8 Juvenile sharks												
9 Narrownose smooth-hound												
10 Benthic-demersal fish II												
11 Flounders												
12 Benthic-demersal fish I	1.E+00	2.E+00	5.E+04	5.E+04	1.E+00							
13 Plownose chimaera	2.E+00											
14 Large hake												
15 Medium-sized hake		2.E+00	1.E+10									
16 Juvenile hake		2.E+00	2.E+01	1.E+01	1.E+00	1.E+10						
17 Pink cusk-eel					1.E+00							
18 Patagonian hoki					1.E+00	2.E+00						
19 Medium-sized pelagic fishes	2.E+00	2.E+00			1.E+00	1.E+00	1.E+00					
20 Small pelagic fishes	3.E+02	2.E+00	5.E+04	5.E+04	1.E+00	3.E+02	3.E+02	2.E+00	1.E+00			
21 Squids	2.E+00	2.E+00	5.E+04	2.E+00	1.E+00	1.E+00	2.E+00		2.E+00			
22 Benthic organisms	2.E+00	2.E+00	1.E+01	2.E+00	1.E+00	1.E+00	1.E+00	1.E+00	1.E+00	1.E+00		
23 Jellyfishes					1.E+00	2.E+00	1.E+00	1.E+00	2.E+00		2.E+00	
24 Zooplankton		2.E+00	1.E+01	1.E+00	1.E+00	1.E+10	1.E+00	1.E+04	1.E+00		1.E+00	2.E+00
25 Phytoplankton						1.E+10		1.E+00			2.E+00	1.E+00
26 Detritus										2.E+00	2.E+00	4.E+00